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Optimal LCL-filter design for a single-phase grid-connected inverter using metaheuristic algorithms☆

Danish Khan [a], Mohammed Qais [b], Irfan Sami [c], Pengfei Hu [a,*], Keping Zhu [d], Almoataz Y. Abdelaziz [e,1]

[a] School of Electrical Engineering, Zhejiang University, Hangzhou 310027, PR China
[b] Electrical Engineering Department, Sana'a University, Sana'a, Yemen
[c] Research and Development Center, Milim Syscon Co., Ltd, Seongnam, South Korea
[d] State Grid Zhejiang Electric Power Company Limited, Economic, and Technical Research Institute, PR China
[e] Faculty of Engineering and Technology, Future University in Egypt, Cairo 11835, Egypt

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ABSTRACT

The inductor-capacitor-inductor (LCL) filter is used to lower the high-frequency switching noise of a grid-connected inverter (GCI). However, a robust design of the LCL filter is a challenge due to its complex model, variations in the operating conditions of the grid, and its stability gain margin. Therefore, this paper designs an optimal LCL filter for GCI and considers the changing grid conditions. This paper utilized four metaheuristic optimization algorithms, including the Whale Optimization Algorithm, the Circle Search Algorithm, the Particle Swarm Optimization, and the gray Wolf Optimization. These algorithms are applied to minimize the total harmonic distortion (THD) and the error between the reference and real grid current. The robustness of the optimal LCL filter is investigated under transient and steady states of the grid by using MATLAB/SIMULINK and real-time simulations. The obtained results revealed that the optimal LCL filter is capable of reducing the THD under different grid variations.

1. Introduction

LCL filters are frequently utilized in distributed power generation systems due to their excellent high-frequency (HF) attenuation characteristics, smaller footprints, and reasonable prices. A precise and dependable LCL filter design is crucial for cost savings, improved filtering performance, and grid-injected current quality [1,2].

Among the design approaches, the parameters of the LCL filter are conventionally selected by the trial-and-error technique. First, the limits of each parameter are determined by constraints such as total harmonic distortion (THD), current ripples [3], and reactive elements [4]. Subsequently, a set of LCL-filter parameters is selected based on real-world engineering knowledge of the application and applied to the real system. If the required grid-integration specifications are not satisfied, the LCL-filter parameters are changed and reevaluated until the design specifications are satisfied [5–7]. Recently, the enhanced trial-and-error method has been investigated by

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* Correponding author.

[1] **Almoataz Y. Abdelaziz** received a Ph.D. in electrical engineering according to the channel system between Ain Shams University, Egypt, and Brunel University, U.K., in 1996. He is currently delegated to the Faculty of Engineering and Technology, Future University in Egypt. His-research areas include the applications of AI, evolutionary, and heuristic optimization techniques to power system planning, operation, and control.

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numerous researchers. For example, in [8], the author presented an optimum parameter for the switching frequency to resonant frequency ratio under a wide range of design constraints. To decrease the capacitance of the filter, a novel reactive power-based design approach was proposed in [9]. References [10] and [11] suggested some new and improved constraints. It is obvious that the trial-and-error approach effectively narrows the set of possible values for the filter parameters. However, since the filter parameters in each validation procedure are created based on experience, the final parameter value may be determined after multiple validations. Because of this, the trial-and-error method makes it hard to predict how well an LCL design will work. With this method, the LCL-filter parameters are just a collection of solutions that pass the verification conditions, so they may not always be the best possible parameters [12].

Designing an LCL filter can be approached using more advanced techniques beyond the conventional parameter selection method. One of these is the enumeration method offers an alternative approach to designing the parameters of an LCL filter. This method fails to find the ideal filter parameters because of the restrictions imposed by the specified LCL parameters [13]. Alternatively, the graphic method may also be used for designing LCL-filter parameters [14]. It is based on drawing an interaction curve between filter parameters and important factors like damping capacity, dimensions, and costs. The filter parameters are determined after determining the values of these variables according to the performance requirements. This approach eliminates the need for trial-and-error but at the expense of a more complicated process.

The aforementioned methods also suffer from several other serious drawbacks. First, the design process does not include a thorough investigation of the damping factor, which is needed for the LCL-filter resonance frequency. Second, IEEE 519 [15] and IEEE 1547 [16] limit the THD of the grid-injected current and the harmonic content for specific frequency ranges, as shown in Table 1. However, the characteristics of the current injected into the grid fall short of grid-connection standards because most of the literature focuses just on the current THD for the LCL-filter parameter design specifications [1]. Also, the error between the reference and measured currents is overlooked; minimizing this error is crucial in avoiding undesirable effects such as increased harmonic distortion, reduced power quality, and added stress on system components. Finally, to lower the cost of the LCL filter, the selected parameters and damping loop must function properly under weak-grid conditions and must withstand any change in the reference current. Nevertheless, none of the preceding studies have taken these aspects into account.

Over the last several years, numerous metaheuristic algorithms (MHAs) have emerged and have proven effective in solving a diverse array of engineering problems [17–19]. These methods are better than traditional ones because they use different ways to search, such as evolutionary computation, swarm intelligence, and memetic hybridization. In this respect, references to the gray Wolf Optimizer (GWO) [20], the Particle Swarm Optimization (PSO) Algorithm [21], the Whale Optimization Algorithm (WOA) [20], and the Circle Search Optimization Algorithm (CSA) [22] stand out. Because of their superiority, effectiveness, and accessibility to many new variables, these algorithms are chosen for designing optimal parameters of LCL-filtered grid-connected inverter (LCL-GCI) systems.

By the above-mentioned studies, this paper aims to describe the design approach for the LCL filter parameters, the damping coefficient of the capacitor current feedback active damping (CCF-AD) method, and the gains of the proportional resonant (PR) controller for the grid-connected inverter (GCI). The objective is to enhance the LCL filter's performance by optimizing its parameters to attenuate the HF harmonics more effectively while minimizing the cost. Further, this paper also explores the robustness, transient behavior, and dynamic response of the LCL-GCI under different conditions, such as grid short circuits and changes in load or reference current. The optimization process aims to ensure that the LCL filter satisfies all technical requirements and conforms to IEEE 519 standards for harmonic content in the grid-injected current across multiple frequency ranges. The design process is achieved using four different metaheuristic techniques. Consequently, a comparative examination of the chosen MHAs has been suggested and will be presented in the subsequent sections. The Simulink/MATLAB software is used for the simulation.

The rest of the paper is structured into the following sections:

The modeling of LCL-filtered single-phase GCI is thoroughly explained in Section 2. Section 3 provides a detailed description of the harmonic-current analysis of a single-phase LCL-GCI. The constraints and objective functions of the LCL filter are explained in Section 4. In Section 5, the selected MHAs are described along with their mathematical representation. In Section 6, the simulation and experimental outcomes obtained under various grid conditions are demonstrated. Additionally, the performance of the GCI system is evaluated under the influence of the simulated MHAs. Finally, the concluding remarks are described in Section 7.

2. LCL-Filter modeling and system description

Fig. 1(a) displays a grid-tied LCL-type single-phase voltage-source inverter (VSI) system. The VSI is energized by a renewable energy source linked to the input side in the form of a DC power source. The inverter generates an output ac voltage(v_i), which is then fed to the LCL filter to reduce the inverter current ripple. The LCL filter contains an inverter-side inductor (L_1), a capacitor (C), and a grid-side inductor(L_2). The sign i_1 signifies the inverter-side current, i_2 represents the grid-side current, and i_c is the filter capacitor

Table 1
Limits on grid-injected current harmonics according to IEEE-519 [15].

Harmonic order h (Odd harmonics)	$h < 11$	$11 \leq h < 17$	$17 \leq h < 23$	$23 \leq h < 35$	$35 \leq h$	THD
Ratio (%) of the rated grid current	4.00	2.00	1.50	0.60	0.30	5.00

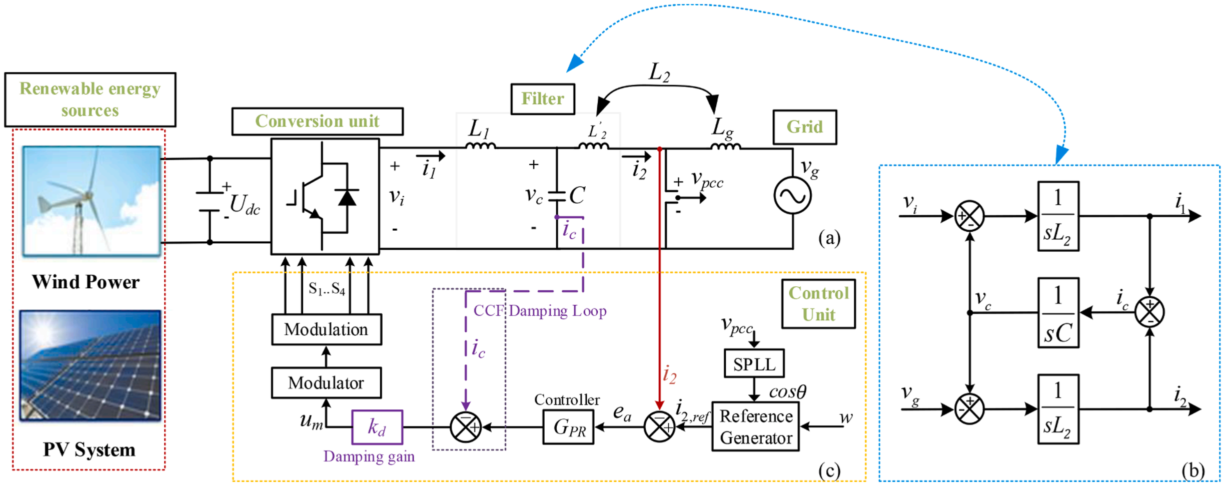

Fig. 1. (a) Schematic diagram of LCL-GCI (b) basic block diagram of LCL-GCI (c) control methodology with CCF-AD loop.

current. Since the grid voltage (v_g) is assumed to be equal to the PCC voltage (v_{pcc}), the grid inductance(L_g) is combined with the grid-side inductance of the LCL filter intoL_2 for simplicity. To simplify the stability analysis and consider a worst-case scenario, the equivalent series resistances of the filter components, which provide extra damping, are ignored. According to Fig. 1(b), the equations related to the LCL-GCI are expressed as follows:

$$\begin{aligned} v_i &= L_1\frac{di_1}{dt} + v_c, v_c = L_2\frac{di_2}{dt} + v_g \\ i_1 &= i_c + i_2, i_c = C\frac{dv_c}{dt} \end{aligned} \tag{1}$$

Ultimately, the control methodology is depicted in Fig. 1(c). It consists of an inner control loop and an active damping technique employed to dampen the resonance peak of the LCL filter. This work selects the conventional CCF-AD to dampen the inherent resonance peak of the LCL filter. In the inner loop of Fig. 1(c), the capacitor current is feedback with a gain (k_d) added to the output of the PR regulator to generate the modulation signal for the single-phase inverter. The introduced k_d in the system can be seen as a virtual damping resistor that operates in conjunction with the filter capacitor. The impact of the CCF-AD method on the LCL-GCI system can be observed in Fig. 2, where the system exhibits a high resonance peak and a -180° phase shift at resonance frequency [1]. However, when the CCF-AD is applied, the LCL-filter resonance is effectively damped, resulting in high stability margins. Unlike a physical damping resistor, the CCF-AD method effectively resolves resonance issues without incurring additional power losses. However, it is important to note that the CCF-AD method is susceptible to uncertainties in system parameters, and the overall stability of the system

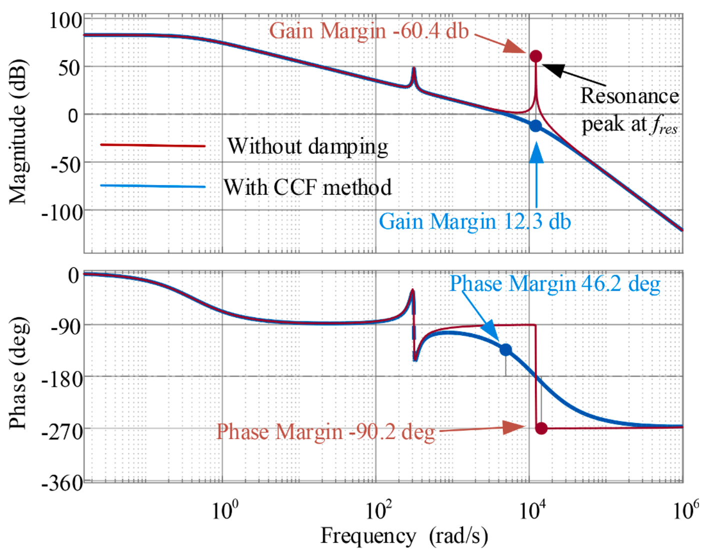

Fig. 2. Frequency response of LCL-GCI system.

can be greatly influenced by the time delay (T_d) associated with digital controllers [23]. This delay refers to the time gap between the sampling instant of input variables and the subsequent updating instant of the control output. It is a crucial aspect to take into account as it can impact the stability and performance of the system. In the context of this study, the time delay T_d corresponds to the sampling period (T_s).

3. Optimal LCL-filter design methodology

3.1. Range of control parameters

Fig. 3(a) depicts a mixed-domain average switching model (ASM) of the LCL-GCI with a CCF-AD loop. The terms, K_{PWM} and e^{-sTs}, represent the converter gain and computation delay, respectively. Considering the grid voltage as a pure sinusoidal waveform at higher frequencies, the ASM represented in Fig. 3(a) can be converted into the configuration illustrated in Fig. 3(b). In this simplified diagram, $G_2(s)$ represents the transfer function from the current $i_2(s)$ to the voltage$v_i(s)$, while $H_c(s)$ represents the transfer function from the current $i_c(s)$ to the voltage $v_i(s)$. The specific mathematical expressions for $G_2(s)$ and $H_c(s)$ are provided by the following equations.

$$G_2(s) = \frac{i_2(s)}{v_i(s)} = \frac{1}{(L_1 + L_2)s}\frac{\omega_{res}^2}{(s^2 + \omega_{res}^2)} \tag{2}$$

$$H_c(s) = \frac{i_c(s)}{v_i(s)} = \frac{s}{L_1(s^2 + \omega_{res}^2)} \tag{3}$$

where ω_{res} represents the angular resonance frequency and is defined as

$$\omega_{res} = \sqrt{\frac{L_1 + L_2}{L_1 L_2 C}} \text{or} f_{res} = \frac{1}{2\pi}.\sqrt{\frac{L_1 + L_2}{L_1 L_2 C}}$$

To analyze the stability of the system in the discrete domain, the zero-order hold (ZOH) method is utilized to obtain the discretized forms of (2) and (3), which are given by (4) and (5) respectively:

$$G_2(z) = \frac{K_{PWM}}{(L_1 + L_2)}\left(\frac{T_s}{z-1} - \frac{z-1}{\omega_{res}}\left(\frac{\sin(T_s\omega_{res})}{z^2 - 2z\cos(T_s\omega_{res}) + 1}\right)\right) \tag{4}$$

$$H_c(z) = \frac{K_{PWM}(z-1)}{\omega_{res}L_1}\left(\frac{\sin(T_s\omega_{res})}{z^2 - 2z\cos(T_s\omega_{res}) + 1}\right) \tag{5}$$

Based on the information provided in Fig. 3(b), (4), and (5), the transfer function $G_{LCL\_d}(z)$ that describes the relationship between the current $i_2(z)$ and the current$i_c(z)$ can be expressed as follows.

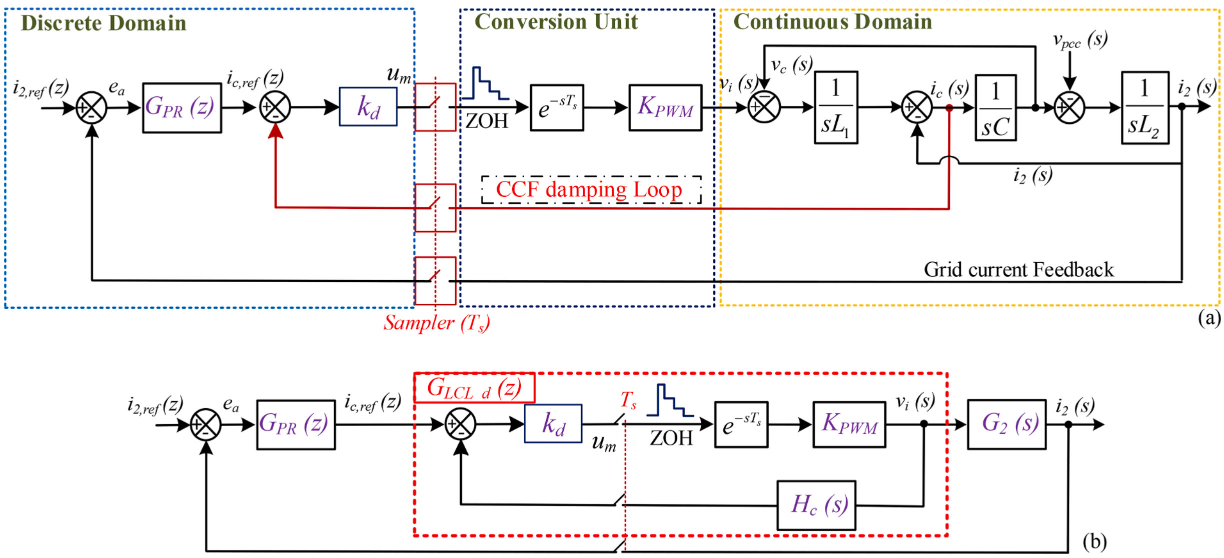

Fig. 3. Mixed-domain ASM of LCL-GCI with CCF-AD loop (b) Simplified equivalent diagram.

$$G_{LCL\_d}(z) = \frac{i_2(z)}{i_c(z)} = \frac{k_d G_2(z)}{1 + k_d H_c(z)} = \frac{ABk_d - Mk_d(z-1)^2}{(z-1)[BR + N(z-1)]} \tag{6}$$

where,

$$A = L_1 T_s K_{PWM}\omega_{res}, B = z^2 - 2z\cos(T_s\omega_{res}) + 1, M = AL_1 K_{PWM}, R = L_1(L_1 + L_2)\omega_{res}, N = A(L_1 + L_2)K_{PWM}k_d.$$

The transfer function of the PR controller in the s-domain and discretized z-domain are presented in (7) and (8), respectively,

$$G_{PR}(s) = K_p + \frac{2K_i\omega_c s}{s^2 + 2\omega_c s + \omega_L{}^2} \tag{7}$$

The parameters ω_c,ω_L,K_p, and K_i refer to the resonant cut-off frequency, fundamental frequency, proportional gain, and resonant gain, respectively.

$$G_{PR}(z) = K_p + \frac{a_1(z^2 - 1)}{b_2 z^2 + b_1 z + b_0} \tag{8}$$

where $a_1 = 4K_i T_s\omega_c, b_2 = T_s{}^2\omega_L^2 + 4T_s\omega_c + 4, b_1 = 2T_s{}^2\omega_L^2 - 8, b_0 = T_s{}^2\omega_L^2 - 4T_s\omega_c + 4$. The discrete z-domain open and closed-loop transfer functions for HF of the LCL-GCI illustrated in Fig. 3(b) can be obtained by utilizing (6) and (8).

$$T_{oL}(z) = G_{PR}(z)G_{LCL\_d}(z) \tag{9}$$

$$F_{cL}(z) = \frac{G_{PR}(z)G_{LCL\_d}(z)}{1 + G_{PR}(z)G_{LCL\_d}(z)}$$

$$F_{cL}(z) = \frac{X_4 z^4 + X_3 z^3 + X_2 z^2 + X_1 z + X_0}{Y_4 z^4 + Y_3 z^3 + Y_2 z^2 + Y_1 z + Y_0} = \frac{N(z)}{D(z)} \tag{10}$$

where,

$$X_4 = -Mk_d a_1, X_3 = Mk_d a_1, X_2 = ABk_d a_1 - Mk_d K_p,$$
$$X_1 = Mk_d K_p - Mk_d a_1, X_0 = Mk_d a_1 + ABk_d a_1 + ABk_d K_p - Mk_d K_p$$

and,

$$Y_4 = b_2 N - Mk_d a_1, Y_3 = b_1 BR + b_2 BR + Mk_d a_1, Y_2 = b_0 N + b_1(BR - 2N) - b_2(BR + N) + ABk_d a_1 - Mk_d K_p,$$
$$Y_1 = b_0(BR - 2N) - b_1(BR + N) - Mk_d K_p + Mk_d a_1, Y_0 = b_0(BR - N) + Mk_d a_1 + ABk_d a_1 + ABk_d K_p - Mk_d K_p$$

The system analysis reveals that the stability of the system primarily depends on the values of K_p and k_d, while the integral gain K_i has minimal impact on stability. K_i is primarily responsible for eliminating steady-state error at the fundamental frequency. Following the Jury Criterion, satisfying the conditions specified in (11) ensures that all the roots of D(z) = 0 lie inside the unit circle. These criteria serve as the foundation for establishing the optimal ranges of PR controller and damping parameters.

$$\begin{aligned} &1.D(1) > 0, 2.D(-1) > 0, 3.|Y_0| < Y_4, 4.\left|\left(Y_0^2 - Y_4^2\right) - (Y_4(Y_0 - Y_1))\right| > 0, \\ &5.|\left(Y_0^2 - Y_4^2\right)^2 - (Y_0Y_4 - Y_1Y_4)^2 - (Y_0Y_3 - Y_1Y_3)(Y_0Y_1 - Y_3Y_4)| > 0. \end{aligned} \tag{11}$$

It should be noted that the conditions mentioned in (11) are not sufficient to determine an optimal range for the control parameters. Additional constraints such as steady-state error, cross-over frequency, gain margin, and phase margin should also be considered when deriving an optimum range for the values of K_p, K_i, and k_d. While this paper emphasizes the analysis of steady-state error, the considerations of other constraints are not discussed in detail but can be obtained from the equations provided in [24].

3.2. Optimal range for steady-state error reduction

The PR controller is employed to minimize the steady-state error of the injected grid currenti_2. However, it is imperative to note that the PR controller alone cannot remove the steady-state error, particularly at the fundamental frequencyf_L. Therefore, when tuning the controller parameters, it is crucial to take into account the steady-state error of i_2, which comprises both an amplitude errore_a and a phase errorp_e. In this study, we will focus only on the amplitude errore_a of the grid current, and its expression with $|T_{fL}|$ representing the gain of the open-loop transfer function at the fundamental frequencyf_L is provided as follows.

$$e_a \approx 10^{-1/20|T_{fL}|} \tag{12}$$

The transfer function $G_{LCL\_d}(s)$ in the continuous s-domain, obtained from (2) and (3) can be expressed as follows.

$$G_{LCL\_d}(s) = \frac{i_2(s)}{i_c(s)} = \frac{K_{PWM}k_d e^{-sT_s}}{L_1L_2Cs^3 + K_{PWM}k_dL_2Ce^{-sT_s}s^2 + (L_1+L_2)s} \tag{13}$$

By considering Fig. 3(b) and utilizing (13), the open-loop transfer function of the LCL-GCI in the s-domain can be derived as shown in (14).

$$T_{oL}(s) = \frac{i_2(s)}{i_{2\_ref}(s)} = \frac{K_{PWM}k_dG_{PR}(s)e^{-sT_s}}{L_1L_2Cs^3 + K_{PWM}k_dL_2Ce^{-sT_s}s^2 + (L_1+L_2)s} \tag{14}$$

At the fundamental frequency f_L, the PR controller equation can be expressed as follows.

$$G_{PR}(j\omega_L) = K_p + \frac{2K_i\omega_c(j\omega_L)}{(j\omega_L)^2 + 2\omega_c(j\omega_L) + {\omega_L}^2} \\ = K_p + K_i = K \tag{15}$$

By ignoring the impact of the capacitor C at f_L in (14) and inserting (15) into (14), the open-loop gain$|T_{fL}|$ of the LCL-GCI can be derived as follows.

$$|T_{fL}| = T_{oL}(j\omega_L) = \frac{K_{PWM}k_dK}{(L_1+L_2)\omega_L} \tag{16}$$

Finally, considering (12) and (16), the amplitude error e_a can be expressed as follows.

$$|T_{fL}| = T_{oL}(f_L) = \frac{K_{PWM}k_dK}{(L_1+L_2)\omega_L} = 20log_{10}\left(\frac{1}{e_a}\right) \tag{17}$$

Based on (17), the gain $|T_{fL}|$ is directly proportional to K_p, $K_{i,}$ and k_d. Therefore, for a given range of k_d, the selection of the controller gains K can be determined based on the desired steady-state error requirement. To guarantee an amplitude errore_a that is below 5% when operating at the rated power, it is necessary for $|T_{fL}|$ to be greater than 30 dB. Hence, to satisfy this constraint even under significant variations in grid impedance, Kshould be greater than the expression provided in (18).

$$K > \frac{30(L_1+L_2)\omega_L}{K_{PWM}k_d} \tag{18}$$

The aforementioned analysis serves as the foundation for determining the feasible ranges of the controller and damping parameters. The specific objective function will be addressed in a subsequent section, followed by the determination of the gain values in Section 6 of this study.

3.3. Harmonic-Current analysis of single phase lcl-gci

The LCL model depicted in Fig. 1(b) can be transformed into its harmonic representation, as illustrated in Fig. 4. The bipolar sinusoidal pulse width modulation method is utilized to produce the modulation signal for GCI switches. In this representation, the angular frequency ω_h corresponds to the harmonic frequencies $[(\omega_{sw}/\omega_L).m + n]th$, where $m = 1, 2, 3$, and so on. The transfer function of $G_2(s)$, as given by (2), can be equivalently expressed in its harmonic form, as shown in (19) [1].

$$|G_2(j\omega_h)| = \left|\frac{i_2(j\omega_h)}{v_i(j\omega_h)}\right| = \frac{1}{L_1L_2C(j\omega_h)^3 + (L_1+L_2)j\omega_h} \tag{19}$$

The following equation can be used to determine the $[(\omega_{sw}/\omega_L).m + n]th$ harmonic current's amplitude [1]:

$$|i_2(j\omega_h)| = \frac{|v_i(j\omega_h)|}{|(L_1+L_2)\omega_h - L_1L_2C{\omega_h}^3|} \tag{20}$$

Hence, the harmonic distortion rate of the $[(\omega_{sw}/\omega_L).m + n]th$ harmonic component of the grid-injected current can be calculated using the following expression.

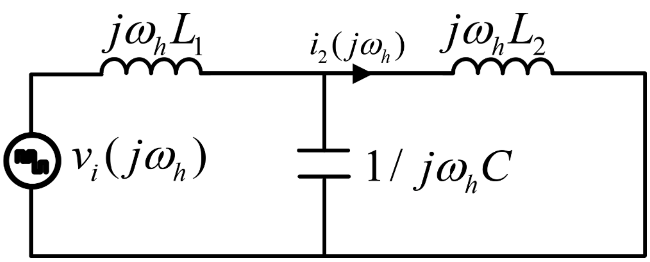

Fig. 4. Harmonic equivalent representation of the LCL-GCI.

$$\begin{aligned} THD_{i2\_hf} &= \frac{i_2(j\omega_h)}{i_{2,1}} \times 100\% \\ &= \frac{|v_i(j\omega_h)|}{i_{2,1}\left|(L_1+L_2)\omega_h - L_1L_2C\omega_h{}^3\right|} \times 100\% \end{aligned} \tag{21}$$

where $i_{2,1}$ represents the amplitude of the fundamental component of grid-injected current.

4. Optimal LCL-filter design methodology

4.1. Objective function

While developing the optimization procedure and objective functions of the LCL-filter parameters, the following factors need to be considered.

1 It is feasible to drastically reduce the low-frequency harmonic components of the grid-injected current by suitably tuning the current controller [25]. This is because the current loop mostly regulates and processes low-frequency digital signals owing to the current controller bandwidth restrictions.
2 One of the challenging criteria to fulfill is the distortion rate of HF components in the grid-injected current. Table 1 demonstrates that if the distortion rate of HF components exceeds 3%, the system fails to meet the IEEE-519 standards, regardless of the THD being below 5%. Therefore, it is imperative to give careful consideration to the HF components alongside THD to ensure system stability following IEEE standards. Therefore, this paper highlights the importance of addressing HF harmonic distortion when designing LCL-filter parameters. The optimization procedure and objective functions consider multiple factors, including THD, total inductance, resonance frequency, controller gains, and damping gain.

4.1.1. Objective I

Combining (21), the first objective of the LCL parameter can be formulated as:

$$\sum THD_{i2\_hf} \leq THD_{TH} \tag{22}$$

Among these, THD_{i2-hf} corresponds to the THD rate of the main HF harmonics present in the grid-connected current, including the harmonics at the switching frequency, its multiples, and sidebands. While THD_{TH} represents the prescribed upper limit for the cumulative HF harmonic distortion rate of the grid-injected current, as specified in Table 1, with THD_{TH} set at 0.3%.

4.1.2. Objective II

Minimizing the steady-state error between the reference and measured current in LCL-GCI applications is crucial for accurate and efficient system operation. From Fig. 3(a) the error between the reference current $i_{2,ref}$ and the measured current i_2 can be written as:

$$e_a(t) = i_{2,ref}(t) - i_2(t) \tag{23}$$

The objective function for reducing the square error between the reference current and measured current can be written as:

$$J = \int_0^T e_a(t)^2 dt \tag{24}$$

where the integral is taken over a time interval. To minimize the objective function, optimization algorithms can be employed to search for the optimal PR controller gains that yield the lowest value of the objective function. Therefore, the fitness function is a multi-objective function as given in (21). This fitness function is subjected to many constraints as in (26)-(31).

$$\min f(x_1, x_2 \ldots x_n) = \min\left(\sum THD_{i2\_hf} + J\right) \tag{25}$$

4.2. Constraints

In this work, the optimization variables in the objective function $\mathbf{X} = [x_1, x_2 \ldots x_n]$ are L_1, C, L_2, K_p, K_i, and k_d of the CCF-AD. However, these variables are subjected to constraints and should be considered in optimization.

$$L_1 + L_2 \leq L_{\max} \tag{26}$$

$$L_{\max} = \frac{3U_{gm}\left(\frac{U_{dc}}{\sqrt{3}} - U_{gm}\right)}{\omega_L P_N} \tag{27}$$

Among them, U_{gm} is the grid phase-voltage peak value, and P_N is the rated reactive power of the system. Moreover, to ensure proper operation, the amplitude of L_1 ripple current $I_{L,1,\ peak}$ should be kept below 20% of the peak value of the grid-injected current I_{gm} [1].

$$L_{1,\min} \geq \frac{U_{dc}}{2\Delta I_{L1,Peak} f_{sw}} \tag{28}$$

where f_{sw} is the switching frequency of the system. The range for f_{res} can be determined as follows:

$$10f_L < f_{res} < 0.5f_{sw} \tag{29}$$

The maximum capacitance value is determined as described in (30).

$$C_{\max.} < \frac{P_N}{3\omega_L v_{g,rms}^2} \tag{30}$$

As a result, the possible range for the capacitor is $C \in (0, C_{max.})$. These constraints are well explained in [14,21].

5. Metaheuristic optimization algorithms

Four MHAs are employed to optimize the LCL filter design for single-phase GCIs. These techniques are the WOA, PSO, CSO, and GWO. These algorithms are detailed in the following.

5.1. Whale optimization algorithm

WOA is an innovative metaheuristic optimization algorithm that was inspired by the way humpback whales hunt with a spiral bubble net to catch food. The research whales randomly follow their prey relative to their positions. This behavior is based on the spiral update location and shrinking encircling technique as stated in [20] and given by the following equations:

$$X_{t+1} = X_{rand} - A.D_{rand} \text{if} |A| > 1 \text{ where} D_{rand} = |C.X_{rand} - X_t| \tag{31}$$

$$X_{t+1} = X^* - A.D \text{if} |A| < 1 \text{ where} D = |C.X^* - X_t| \tag{32}$$

where

$$\begin{array}{l} A = 2.a.r - 2, a = 2 - 2(t/t_{\max}), and C = 2.r \\ X_{t+1} = X^* + D^{'}.e^{bl}.\cos(2\pi l) \end{array} \tag{33}$$

where

$$D^{'} = |X^* - X_t|, l = (a_2 - 1) * rand + 1, \text{and} a_2 = -1 - t/t_{\max}$$

where t is the current state, t_{max} is the maximum number of states, b is a constant and set to 1, and r is a real random number between 0 and 1.

5.2. Gray wolf optimization

GWO was created to resemble the social structure and hunting techniques of gray wolves [20]. The GWO uses four different wolf populations and the best options are expected to be the α wolves that engage in hunting. The β wolves and δ wolves are the second and third best choices, respectively, which might assist the wolves while they deliberate about which option to choose. The ω wolves are the followers.

$$\vec{D}_\alpha = \left|\vec{C}_1.\vec{X}_{\alpha t} - \vec{X}_t\right|, \vec{D}_\beta = \left|\vec{C}_2.\vec{X}_{\beta t} - \vec{X}_t\right|, \vec{D}_\delta = \left|\vec{C}_3.\vec{X}_{\delta t} - \vec{X}_t\right| \tag{34}$$

$$\vec{X}_1 = \vec{X}_{\alpha t} - \vec{A}_1.\vec{D}_\alpha, \vec{X}_2 = \vec{X}_{\beta t} - \vec{A}_2.\vec{D}_\beta, \vec{X}_3 = \vec{X}_{\delta t} - \vec{A}_3.\vec{D}_\delta \tag{35}$$

$$\vec{X}_{i+1} = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \tag{36}$$

$$\vec{a} = 2 - 2 \times t/t_{\max} \tag{37}$$

$$\vec{A} = 2\vec{a}.\vec{r}_1 - \vec{a} \tag{38}$$

$$\vec{C} = 2.\vec{r}_2 \tag{39}$$

where r_1 and r_2 are real random numbers between 0 and 1, A is the coefficient vector, X is the wolf position vector, X_α is the prey position vector, and D is the distance between the wolf and prey positions.

5.3. Particle swarm optimization

The PSO involves a search process utilizing a swarm of particles that undergo mutations over time. The aim is to locate the optimal solution by having each particle move towards its previous best position ($\mathbf{P}_t$) and the best position in the entire swarm ($\mathbf{G}_t$) [21]. One has.

$$\mathbf{P}_t = \text{argmin}[f(\mathbf{X}_t)] \tag{40}$$

$$\mathbf{G}_t = \text{argmin}[f(\mathbf{X}_t)] \tag{41}$$

where i stands for the particle index, N_P for the overall particle count, t for the iteration number, P for the position, and f for the fitness function. The location of particle P and velocity V are updated by the following equations:

$$V_{t+1} = \omega.V_t + c_1.r_1(P_t - X_t) + c_2.r_2(G_t - X_t) \tag{42}$$

$$X_{t+1} = X_t + V_{t+1} \tag{43}$$

where a balance between global exploration and local exploitation is provided by the inertia weight ω. The range of the random variables r_1 and r_2 is [0, 1], and their distribution is uniform. On the other hand,c_1 and c_2 are known as "acceleration coefficients," which are positive constant parameters.

5.4. Circle search algorithm

CSA is a new optimization algorithm introduced in [22]. The exploitation and exploration of this algorithm are well explained in [22]. When the center of the circle is used as a reference point, the angle that exists between the tangent line contact point and the circumference of the circle begins to gradually decrease as one approaches the center of the circle. The following describes the process that this algorithm goes through to find the optimal answer.

- Initialization: During the initialization stage of the CSA, it is important to randomize the dimensions of all search agents consistently. Inconsistent randomization can lead to unexpected outputs. The search agents are typically initialized within the upper-bound (UB) and lower-bound (LB) limits of the search space, using (44) as a guideline.

$$X_t = LB + r \times (UB - LB) \tag{44}$$

whereX_t (touching point of each circle) is the search agent of CSA and r the random vector whose position lies within [0, 1].

- Update the search agent position: As stated in (45), the search-agentX_t position is modified based on the estimated optimal position X_c (the center point of the circle and expected to be the best location in the algorithm).

$$X_{t+1} = (X_c - X_t) \times \tan(\theta) + X_c \tag{45}$$

where the angleθis crucial in developing the CSA and may be found by using the formula

$$\theta = \left\{ \begin{matrix} w \times randt > (c \times t_{\max}) \\ w \times potherwise \end{matrix} \right\} \tag{46}$$

$$\omega = \omega \times rand - \omega \tag{47}$$

$$a = \pi - \pi \times \left(\frac{t}{t_{\max}}\right)^2 \tag{48}$$

$$p = 1 - 0.9 \times \left(\frac{t}{t_{\max}}\right)^{0.5} \tag{49}$$

where c indicates the maximum iteration in percentage and *rand* is a random number that varies between 0 and 1. Equation (47) demonstrates that, when the number of repetitions increases, the variable w moves from -π to 0. The variable a varies between π and 0 according to (48), while the variable 'p' varies between 1 and 0 according to (49). Similarly, the angle θ changes between -π and 0.

6. Results and discussion

6.1. Optimization results

The goal of this work is to offer an algorithm that gives the optimum values for the LCL-filter parameters, PR controller gain, and the damping gain of the CCF-AD method so that a high-quality grid-injected current is obtained. Applying the GCI parameters listed in Table 2 to (27), (28), and (31), the ranges obtained for the LCL-filter parameters are $L_1 \in$ (0.2 mH, 4.2 mH), $L_2 \in$ (0 mH, 4.2 mH), and $C \in$ (0 uF, 16 uF). Furthermore, the range of PR controller gains, as well as the limit of the CCF-AD method gain, are determined using (18), while ensuring compliance with the constraints specified in [24].

Figs. 5(a)–(c) show the optimization procedures of L_1, L_2, and C respectively. The optimization procedure of the LCL-filter parameters reaches a steady state for each method when the number of iterations reaches a particular value, indicating that the optimal solution is achieved. Among them, CSA yields lower values for L_1 and L_2 compared to the other algorithms, resulting in decreased inductance values. The optimization outcomes for the LCL parameters are listed in Table 3.

The optimization process, depicted in Figs. 6(a)–(b), achieves the optimal gains for the PR controller over 200 iterations. WOA and CSA methods exhibit faster convergence and yield relatively lower gain values compared to other methods, offering potential advantages in terms of stability margins and reduced sensitivity to parameter variations. The resonant cut-off frequency ω_c is fixed at 3.14 rad/s for all MHAs used in this study. Moreover, determining the value of k_d is vital for achieving proper stability gain margins and cross-over frequency as per the IEEE standard to efficiently suppress the resonance peak in the LCL-GCI system. The optimization procedure for obtaining the optimal value of k_d using the applied metaheuristic optimization algorithms is depicted in Fig. 6(c).

Fig. 7 illustrates the fitness function evolution over time for the four selected MHAs. The graph indicates that the fitness values obtained during the optimization process for all MHAs are below 1, indicating the suitability of the optimized LCL parameters for the initial objectives. Notably, the fitness value of the CSA algorithm is the lowest among the considered MHAs, showcasing its superior performance in terms of both local and global search capabilities. These results highlight the effectiveness of the relatively new CSA algorithm as a global metaheuristic approach for designing LCL-filter parameters in GCI applications.

6.2. Simulation results

To evaluate the performance of the design parameters achieved through the optimization methods, a comparative simulation is conducted between the selected MHAs. In the MATLAB/Simulink environment, an LCL-filtered single-phase grid-tied VSI system like the one shown in Fig. 1(a) is simulated.

Fig. 8 shows the steady-state and transient responses of the current control loop to a 50% step change in the reference current. The results indicate that all selected MHAs effectively track the reference current. However, the CSA algorithm exhibits smaller overshoot and steady-state errors compared to other algorithms, as observed in both the main and zoomed-in sections of Fig. 8(a). Additionally, in Fig. 8(b), when the reference current steps from 15 A to 30 A, the THD of the grid-injected current remains below the acceptable limit of 5% according to IEEE standards. Notably, the CSA algorithm demonstrates the lowest transient overshoot among the tested MHAs, further emphasizing its superior performance.

Fig. 8(c) displays the plot of the RMS grid-injected current under a 50% step change, revealing noticeable oscillations in WOA, GWO, and PSO, while CSA parameters exhibit minimal oscillation. Fig. 8(d) demonstrates the response of active and reactive power to the step change. Active power reaches a steady state within 0.02 s without substantial overshoot, while transient spikes in reactive power remain below 40 VAR. The CSA optimization method shows fewer oscillations compared to other MHAs, ensuring stable system performance.

It is worth noting that relying solely on THD as the primary design criterion for LCL filters, as is often the case in literature, can be problematic in practice. According to recent research [21], while the THD may fall within acceptable limits (less than 5%), the percentage of HF harmonic distortion can still exceed 0.3%, violating grid connection codes as listed in Table 1. Therefore, it is crucial to pay special attention to the HF harmonic distortion rate when determining the LCL filter parameters during the design process. Figs. 9(a)–(d) displays the harmonic spectra of the grid-injected current for the four MHAs under consideration. It is noteworthy that the HF harmonic distortion rate remains below the threshold of 0.3% for each of the MHAs, with the CSA technique yielding a lower value as demonstrated in Fig. 9(d).

To evaluate the robustness, the system is subjected to a short circuit from 0.1 s to 0.13 s, as shown in Fig. 10. During this test, a resistor of 0.25 ohm is used in series with the grid so that the current passes through the short wires, which makes the system more

Table 2
Parameters of GCI.

Parameter	Symbol	Value
DC voltage	U_{DC}	400 V
Rated power	P_N	5 kW
Grid voltage (RMS)	V_g	230 V
Grid frequency	f_L	50 Hz
Sampling frequency	f_s	20 kHz
Switching frequency	f_{sw}	10 kHz

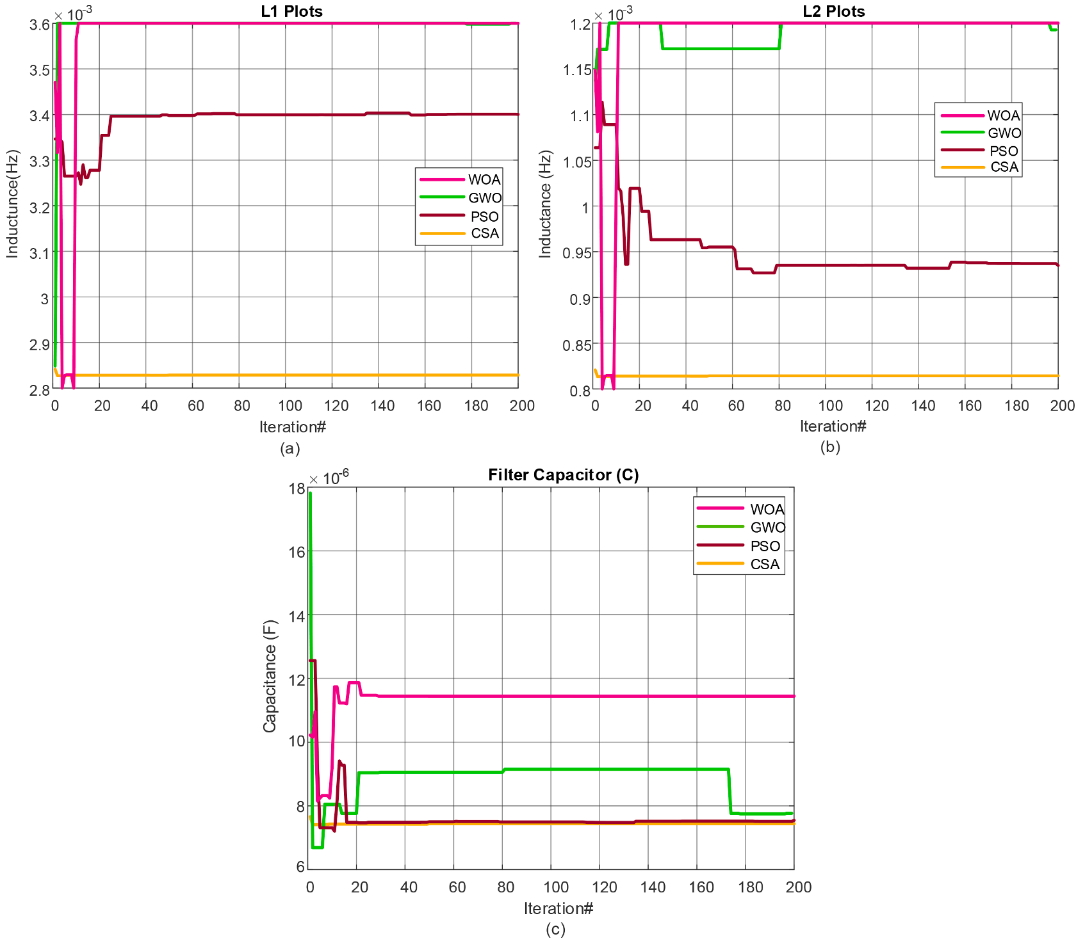

Fig. 5. Optimization procedure and outcomes of the *LCL* parameters under different MHAs (a) L_1 (b) L_2 (c) C.

Table 3
Optimum values of LCL-filter parameters, damping coefficient, and PR controller gain with simulated MHAs.

Algorithm	L_1 (H)	L_2 (H)	C (F)	k_d	K_p	K_r
WOA	0.00343	0.00112	11.4969e-06	37.9832	27.0689	376.245
GWO	0.00360	0.00119	7.87697e-06	19.5447	21.8378	933.397
PSO	0.00340	0.00093	7.54576e-06	30.0063	24.0025	1066.31
CSA	0.00282	0.00081	7.50808e-06	17.1177	17.9217	358.401

challenging. Fig. 10(a) reveals that the grid RMS voltage drops 90% from its nominal value during the short-circuit period. However, after the fault is removed, the grid-injected RMS current shows a smooth transition for all MHAs, with CSA exhibiting reduced oscillation and lower transient spikes during the recovery time, as presented in Fig. 10(b). The influence of the short circuit on the active and reactive power of the system is shown in Fig. 10(c). During the short-circuit period, the active power approaches zero, while the reactive power exhibits transient behavior with oscillations and spikes. These transient spikes are present in all algorithms but are less pronounced in CSA. Increasing the value of inverter-side inductance can mitigate these spikes at the expense of higher filter costs. After removing the short-circuit fault, both active and reactive powers quickly recover to their steady states. The CSA algorithm demonstrates lower oscillations during the recovery process compared to the other algorithms.

6.3. Experimental results

Experimental results were achieved to evaluate the performance of the current control loop for grid-injected current under steady-state and transient conditions. The control strategy is implemented using the Imperix B-Box RCP3.0 and HIL-404 in a Controller-Hardware-in-the-Loop (CHIL) environment. The B-Box RCP3.0 generates a PWM signal from the control block, which is received by Typhoon HIL-404 to drive the grid-connected inverter. Fig. 11 illustrates the relationship between the B-Box RCP3.0 controller and

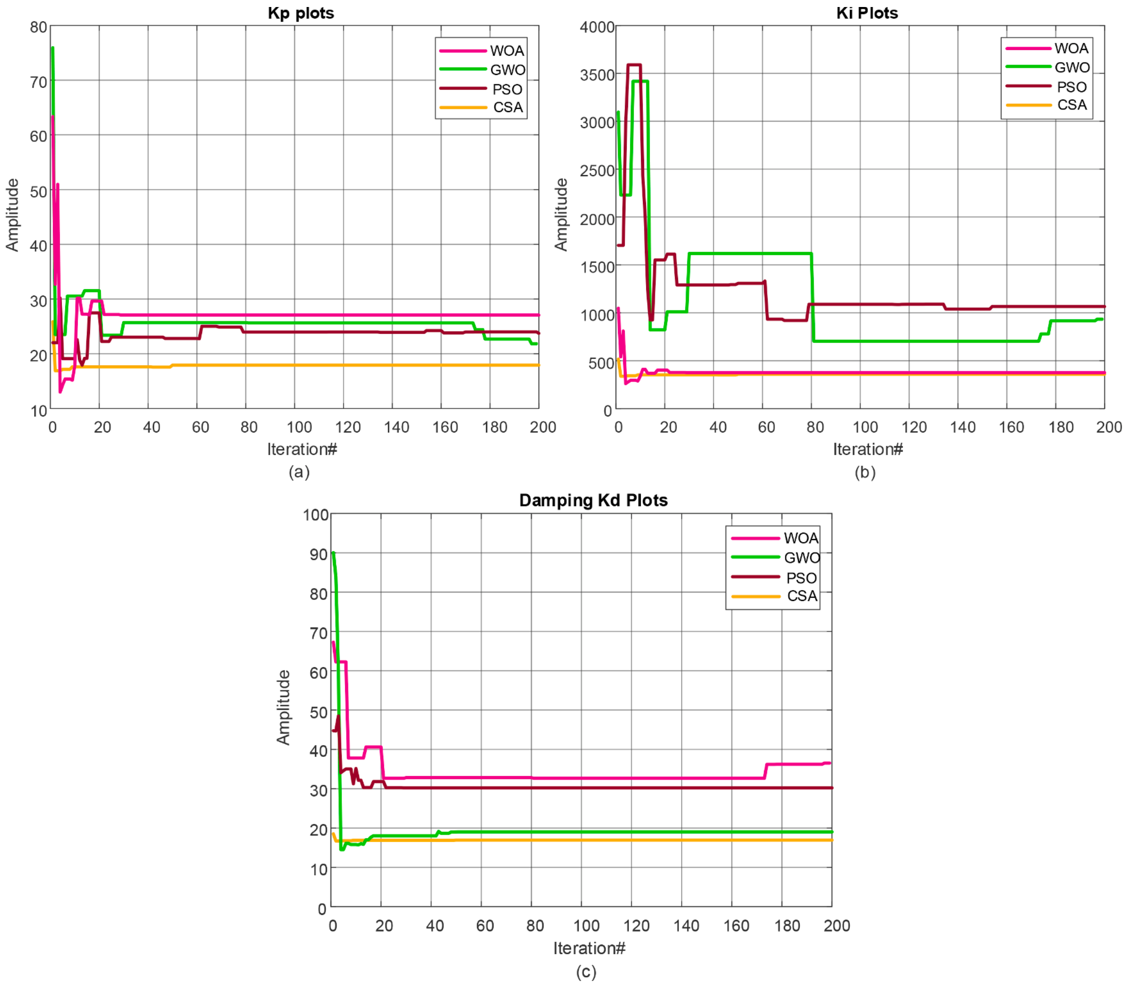

Fig. 6. Optimization procedure of the PR controller gains (a) K_p (b) K_i and damping gain (c) k_d under different MHAs.

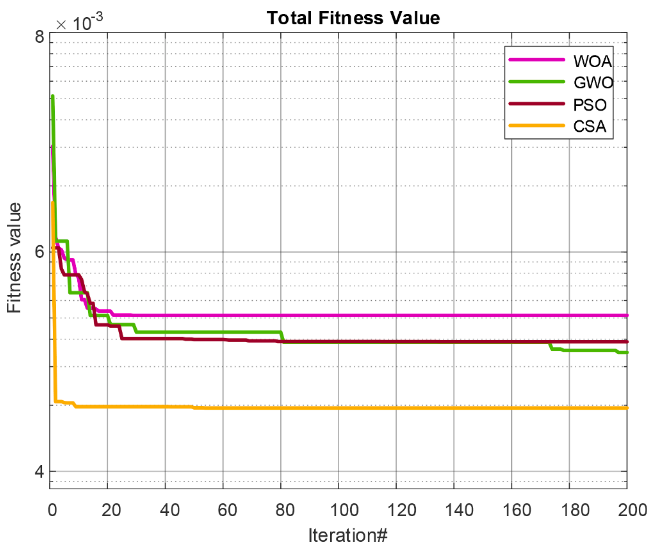

Fig. 7. Convergence curve of objective function under different MHAs.

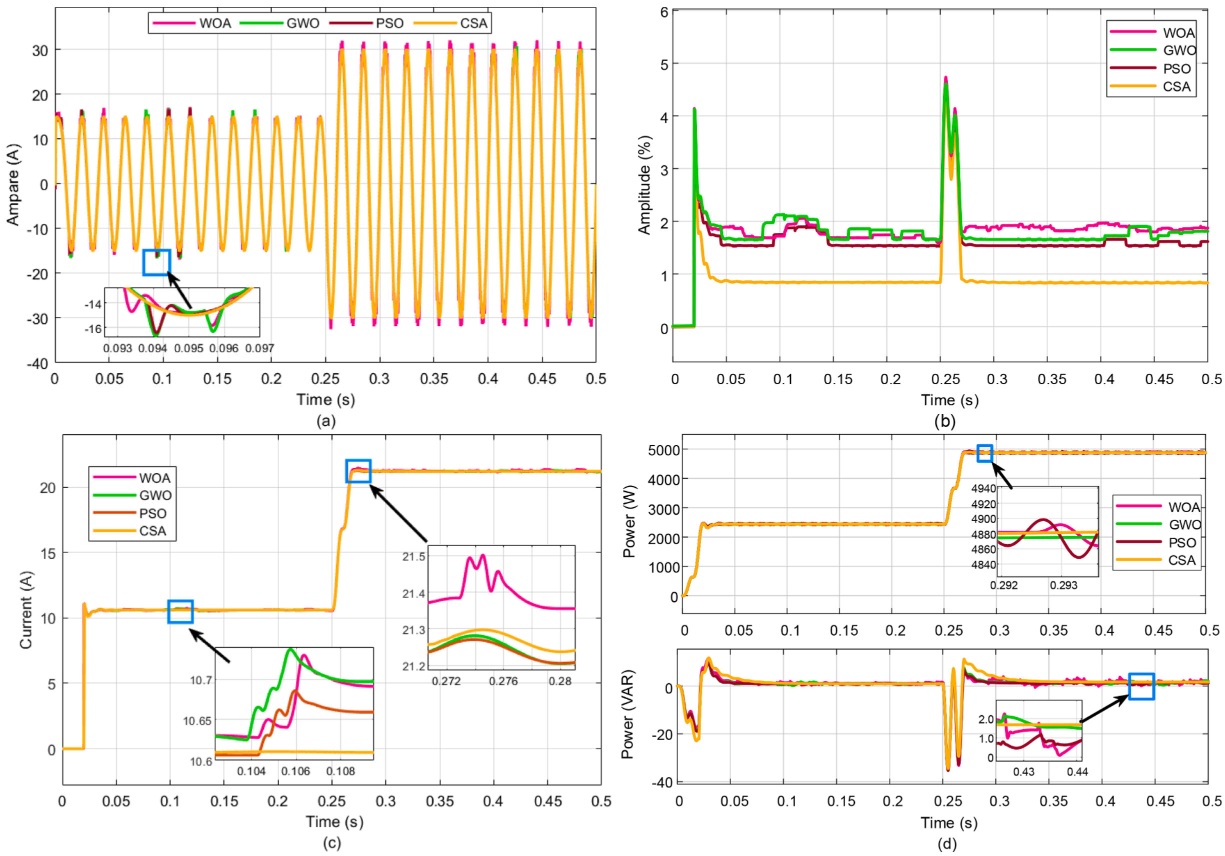

Fig. 8. Simulation results of applied MHAs under the change in reference current. (a) current i_2. (b) THD. (c) RMS current $i_{2,\ RMS}$ (d) reactive and active powers.

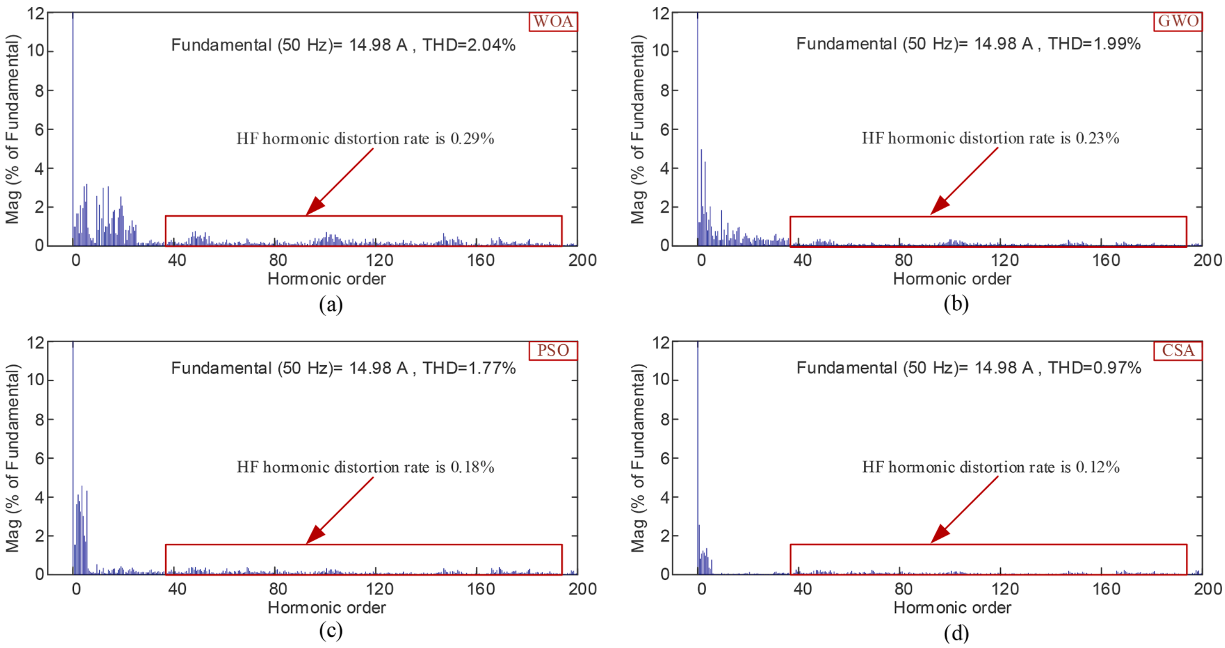

Fig. 9. Harmonic spectra of current i_2 under applied MHAs.

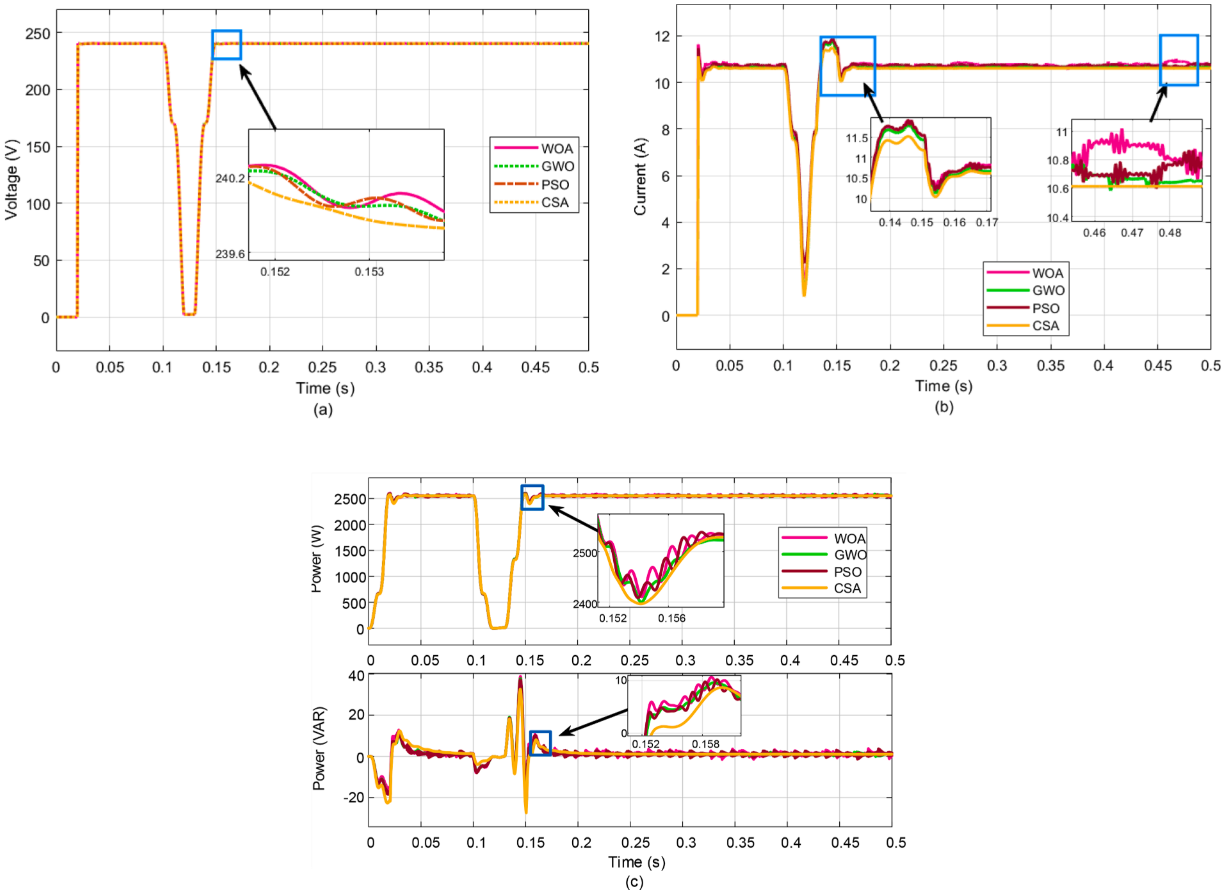

Fig. 10. Simulation outcomes for (a) grid RMS voltage (b) injected RMS grid current (c) active and reactive power under grid short circuit.

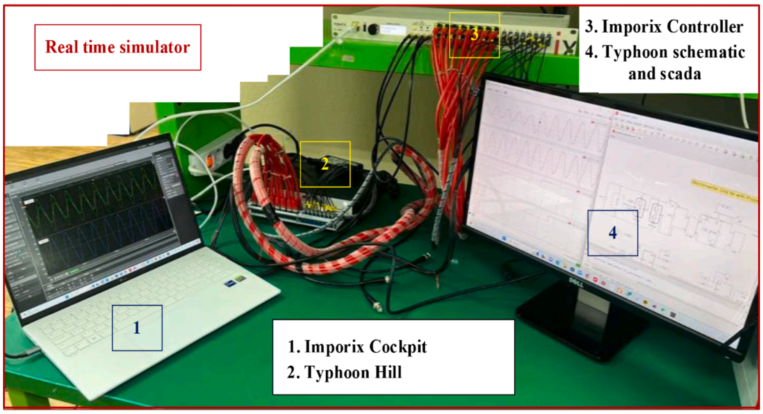

Fig. 11. Experimental setup.

the HIL-404 plant, where the software part of the controller is implemented in MATLAB/Simulink using the B-Box library and Typhoon Schematic hardware. Typhoon SCADA is used for analyzing the hardware dynamics.

To evaluate the transient response, a step change of 50% in the reference current was applied to the system, and the responses of the grid-injected current were observed using the simulated MHAs as presented in Fig. 12. The experimental outcomes in Fig. 12(a) show that the grid-injected current is effectively tracked by the reference current for all the selected MHAs. Moreover, when the reference current was stepped up from 15 A to 30 A at 0.37 s, the THD of the grid-injected current was found to increase to almost 4.5%, as shown in Fig. 12(c), which is still within the acceptable limits.

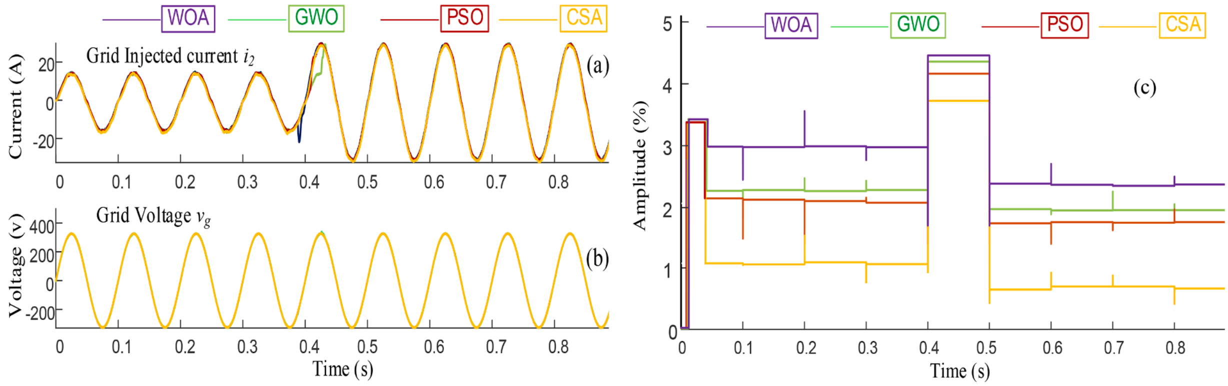

Fig. 12. Experimental outcomes for applied MHAs under the change in reference current. (a) injected current i_2 (b) grid voltage v_g and (c) THD rate of the injected current i_2.

The active and reactive power response of the system to a step change in the reference current is presented in Fig. 13. In Fig. 13(a), the active power demonstrates a gradual increase in response to the increment in the current, settling to a steady state within a short time. Although the active power exhibits some oscillations during the transient period, there is no substantial overshoot. Regarding the reactive power, its magnitude remains below 40 VAr during the transient period, indicating negligible detrimental effects on the overall system performance as presented in Fig. 13(b). To assess the robustness of the LCL-GCI, a short-circuit fault was introduced, and the grid-injected current during the fault was examined, as depicted in Fig. 13(c). The results indicate a smooth transition of the current from the rated value to zero and from zero back to the rated value for all selected MHAs. Notably, the CSA algorithm exhibits a lower transient spike during the recovery time in comparison to the other three algorithms. The experimental results demonstrate that the CSA algorithm outperforms other applied algorithms, and therefore, the findings are consistent with the simulation results.

7. Conclusion

In this work, the metaheuristic optimization algorithms were applied to optimally design the LCL filter components and the parameters of the PR controller and damping gain. The main objective of the optimal design is to improve the quality of the grid-injected current and at the same time enhance the stability of the grid-connected LCL filter. This can be achieved by applying metaheuristic optimization algorithms to identify the optimal components of the LCL filter and the optimal gains of the PR controller and the damping of capacitor current feedback. Four metaheuristic algorithms, including WOA, GWO, PSO, and CSA, are applied to minimize the objective functions considering the constraints of the selection of the optimal variables. A comparative analysis has been done between the applied algorithms to find out which one gives the best THD (according to IEEE 519 standards THD $\leq$ 5%). Based on the simulation and experimental results, the CSA algorithm not only achieved the lowest THD and best power quality but also demonstrated the smallest steady-state error and overshoot. Therefore, in terms of effectiveness, robustness, and stability, the CSA obtained better results compared with the other three algorithms. In particular, the results demonstrate that the optimized filter parameters, PR controller gain, and CCF-AD gain obtained using this algorithm have adequate HF harmonic attenuation capability and satisfy the grid connection codes regarding the HF harmonic content of the grid-injected current. Furthermore, the suggested approach is robust for grid short-circuit conditions and step changes in reference current. Finally, the resulting designed LCL values offer the best balance between LCL filter cost and filtering performance.

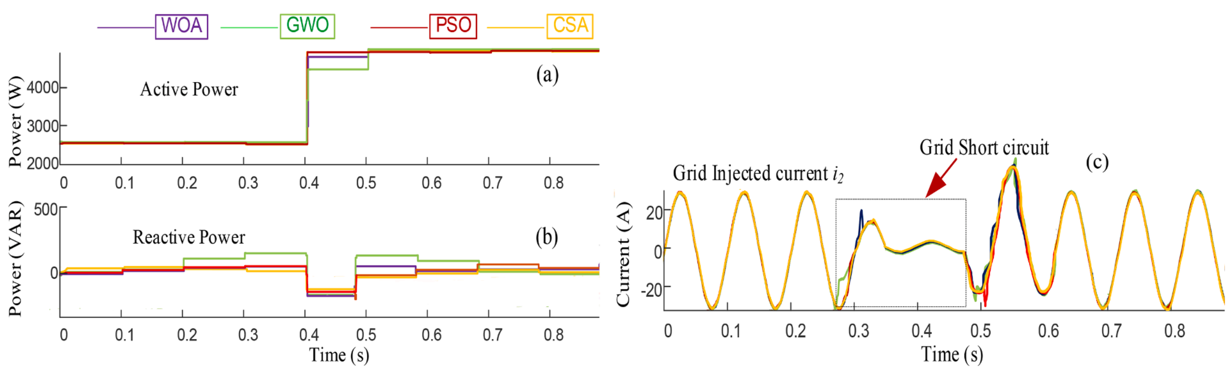

Fig. 13. Experimental outcomes for applied MHAs under the change in reference current. (a) active power (b) reactive power and (c) injected grid current i_2 under a grid short circuit.

Declaration of Competing Interest

"The Author's declares that there is no conflict of interest."

Data availability

Data will be made available on request.

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